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Stewart et al.

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(54) **STRAWBERRY PLANT NAMED ‘DRISSTRAWEIGHTYONE’**

(50) Latin Name: *Fragaria x ananassa*
Varietal Denomination: **DrisStrawEightyOne**

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(52) **U.S. Cl.**
USPC **Plt./209**

(58) **Field of Classification Search**
USPC **Plt./208, 209**
See application file for complete search history.

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(57) **ABSTRACT**

A new and distinct variety of strawberry plant named ‘DrisStrawEightyOne’, particularly selected for the sweetness and firmness of its fruit, is disclosed.

7 Drawing Sheets

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STRAWBERRY PLANT NAMED 'DRISSTRAWEIGHTYONE'

Botanical classification: *Fragaria* x *ananassa*

Varietal denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawEightyOne'.

BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18th century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawEightyOne'.

Strawberry plant variety 'DrisStrawEightyOne' originated from a cross between the proprietary female parent '42T172' (unpatented) and the proprietary male parent '8T145' (unpatented). Progeny plants from this cross of '42T172' x '8T145', including 'DrisStrawEightyOne', were asexually propagated via stolons in Shasta County, Calif., USA in April of 2013. Strawberry plant variety 'DrisStrawEightyOne' was later specifically identified and selected in Monterey County, Calif., USA in May of 2014.

'DrisStrawEightyOne' was subsequently asexually propagated via stolons, and underwent further testing at test plots in Monterey County, Calif., USA for five years (2014 to 2019). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons and tissue culture.

'DrisStrawEightyOne' exhibits the following distinguishing characteristics when grown under normal horticultural practices in Monterey County, Calif., USA:

1. Inflorescence on the same level as foliage;
2. Absent or very weak anthocyanin coloration of stolon;

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3. Serrate to crenate shape of margin of terminal leaflet; and
4. Calyx attachment level with fruit.

'DrisStrawEightyOne' was particularly selected for the sweetness and firmness of its fruit.

DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs which show fruit of the plant, flowers, leaves, and the plants. The colors shown are as true as can be reasonably obtained by conventional photographic procedures. The photographs are of plants that are eleven months old.

FIG. 1 illustrates whole fruit of variety 'DrisStrawEightyOne'.

FIG. 2 illustrates longitudinal sections of fruit of variety 'DrisStrawEightyOne'.

FIG. 3 illustrates the upper surfaces of flowers of variety 'DrisStrawEightyOne'.

FIG. 4 illustrates the lower surfaces of flowers of variety 'DrisStrawEightyOne'.

FIG. 5 illustrates leaves of variety 'DrisStrawEightyOne'.

FIG. 6 illustrates a side view of a plant of variety 'DrisStrawEightyOne'.

FIG. 7 illustrates an aerial view of a plant of variety 'DrisStrawEightyOne'.

DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawEightyOne'. The data which define these characteristics is based on observations taken in Monterey County, Calif., USA from 2014 to 2019. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawEightyOne' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawEightyOne' was taken from plants that were eleven months old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The R.H.S. Colour Chart of The Royal Horticultural Society of London (R.H.S.) (2015 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary*, 2nd edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:

Species.—*Fragaria* x *ananassa*.

Common name.—Strawberry.

Denomination.—'DrisStrawEightyOne'.

Parentage:

Female parent.—Proprietary strawberry plant '42T172' (unpatented).

Male parent.—Proprietary strawberry plant '8T145' (unpatented).

Plant:

Height.—33.0 cm.

Diameter.—59.1 cm.

Number of crowns per plant.—8.8.

Growth habit.—Semi-upright.

Stolon:

Average number of daughter plants per square foot.—22.

Length.—50.6 cm.

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Diameter at bract.—2.8 mm.

Anthocyanin coloration.—Absent or very weak.

Stolon color with anthocyanin coloration.—RHS 144B (Strong yellow-green).

Leaf:

Number of leaflets.—Three only.

Color of upper surface.—RHS NN137A (Greyish olive green).

Color of lower surface.—RHS 138A (Moderate yellowish green).

Variation.—Absent.

Terminal leaflet.—Length: 7.2 cm. Width: 7.4 cm. Length/width ratio: 0.97. Number of teeth/terminal leaflet: 17.9. Shape of base: Rounded. Margin: Serrate to crenate. Shape in cross section: Concave.

Petiole.—Length: 22.1 cm. Diameter: 2.5 mm. Attitude of hairs: Slightly outwards. Bract frequency (number present on each petiole): 1.3. Color: RHS 144C (Strong yellow-green).

Petiolule.—Length: 18.2 mm. Diameter: 1.1 mm. Color: RHS 144C (Strong yellow-green).

Stipule.—Length: 3.9 cm. Width: 18.4 mm. Anthocyanin coloration: Weak. Anthocyanin color: RHS 62C (Light purplish pink). Color: RHS 144D (Light yellow-green).

Inflorescence:

Position in relation to foliage.—Same level.

Pedicel.—Attitude of hairs: Slightly outwards. Length: 18.3 mm. Diameter: 1.9 mm. Color: RHS 144B (Strong yellow-green).

Flower.—Flower diameter (petal tip to petal tip on non-flattened flower): 17.56 mm. Arrangement of petals: Overlapping. Stamen: Present. Typical and observed number of flowers per plant: 6.6.

Petal.—Length: 7.72 mm. Width: 9.05 mm. Length/width ratio: 0.89. Typical and observed petal number: 5. Color of upper side: RHS NN155B (Pinkish white). Color of lower side: RHS 155D (Yellowish white). Base shape: Concavo-convex. Apex shape: Rounded. Margin: Entire.

Calyx.—Diameter (sepal tip to sepal tip, measured on back of flower): 22.97 mm.

Sepal.—Length (sepal tip to point of attachment to receptacle): 8.97 mm. Width: 5.54 mm. Typical and observed sepal number: 9.9. Color of upper surface: RHS 139A (Dark yellow-green). Color of lower surface: RHS 138A (Moderate yellowish green).

Fruit:

Length.—35.8 mm.

Width.—30.9 mm.

Length/width ratio.—1.16.

Fruit hollow length.—18.54 mm.

Fruit hollow width.—8.47 mm.

Fruit hollow length/width ratio.—2.189.

Fruit weight.—21.8 grams.

Shape.—Conical.

Glossiness.—Medium.

Firmness.—Firm.

Sweetness.—Strong.

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Color.—RHS N45A (Moderate red).

Position of achenes.—Level with surface.

Number of achenes per fruit.—187.

Position of calyx attachment.—Level with fruit.

Attitude of sepals.—Outwards.

Color of flesh (excluding core).—RHS 42A (Vivid reddish orange).

Color of core.—RHS 44A (Vivid red).

Production:

Flowering interval.—Mid-February to early November.

Harvest interval.—Mid-March to early November.

Type of bearing.—Partially remontant.

Productivity.—33,102 kg to 52,670 kg of fruit per acre per season from eleven-month-old plants when grown in Monterey County, Calif., USA.

Resistance to abiotic stress, pests, and diseases:

Two-spotted spider mite (tetranychus urticae).—Moderately susceptible.

Botrytis fruit rot (botrytis cinerea).—Moderately susceptible.

Powdery mildew (podosphaera macularis).—Moderately susceptible.

COMPARISON WITH PARENTAL AND COMMERCIAL VARIETIES

‘DrisStrawEightyOne’ differs from the proprietary female parent ‘42T172’ (unpatented) in that fruit of ‘DrisStrawEightyOne’ are sweeter and more prone to raised seeds and uneven coloration than fruit of ‘42T172’.

‘DrisStrawEightyOne’ differs from the proprietary male parent ‘8T145’ (unpatented) in that fruit of ‘DrisStrawEightyOne’ are sweeter and have better shelf life than fruit of ‘8T145’. In addition, ‘DrisStrawEightyOne’ is less susceptible to powdery mildew when compared with ‘8T145’.

‘DrisStrawEightyOne’ differs from the commercial variety ‘DrisStrawFiftyThree’ (U.S. Plant Pat. No. 29,749) in that ‘DrisStrawEightyOne’ has a serrate to crenate margin of terminal leaflet, absent or very weak anthocyanin coloration of stolon, calyx attachment level with fruit, and an outwards attitude of sepals on fruit, whereas ‘DrisStrawFiftyThree’ has a crenate margin of terminal leaflet, strong anthocyanin coloration of stolon, calyx attachment raised from fruit, and an upwards attitude of sepals on fruit.

‘DrisStrawEightyOne’ differs from the commercial variety ‘DrisStrawNine’ (U.S. Plant Pat. No. 20,733) in that ‘DrisStrawEightyOne’ has absent or very weak anthocyanin coloration of stolon, a serrate to crenate margin of terminal leaflet, weak anthocyanin coloration of stipule, and inflorescence on the same level as foliage, whereas ‘DrisStrawNine’ has medium anthocyanin coloration of stolon, a crenate margin of terminal leaflet, absent or very weak anthocyanin coloration of stipule, and inflorescence above foliage.

We claim:

1. A new and distinct variety of strawberry plant named ‘DrisStrawEightyOne’ as shown and described herein.

* * * * *

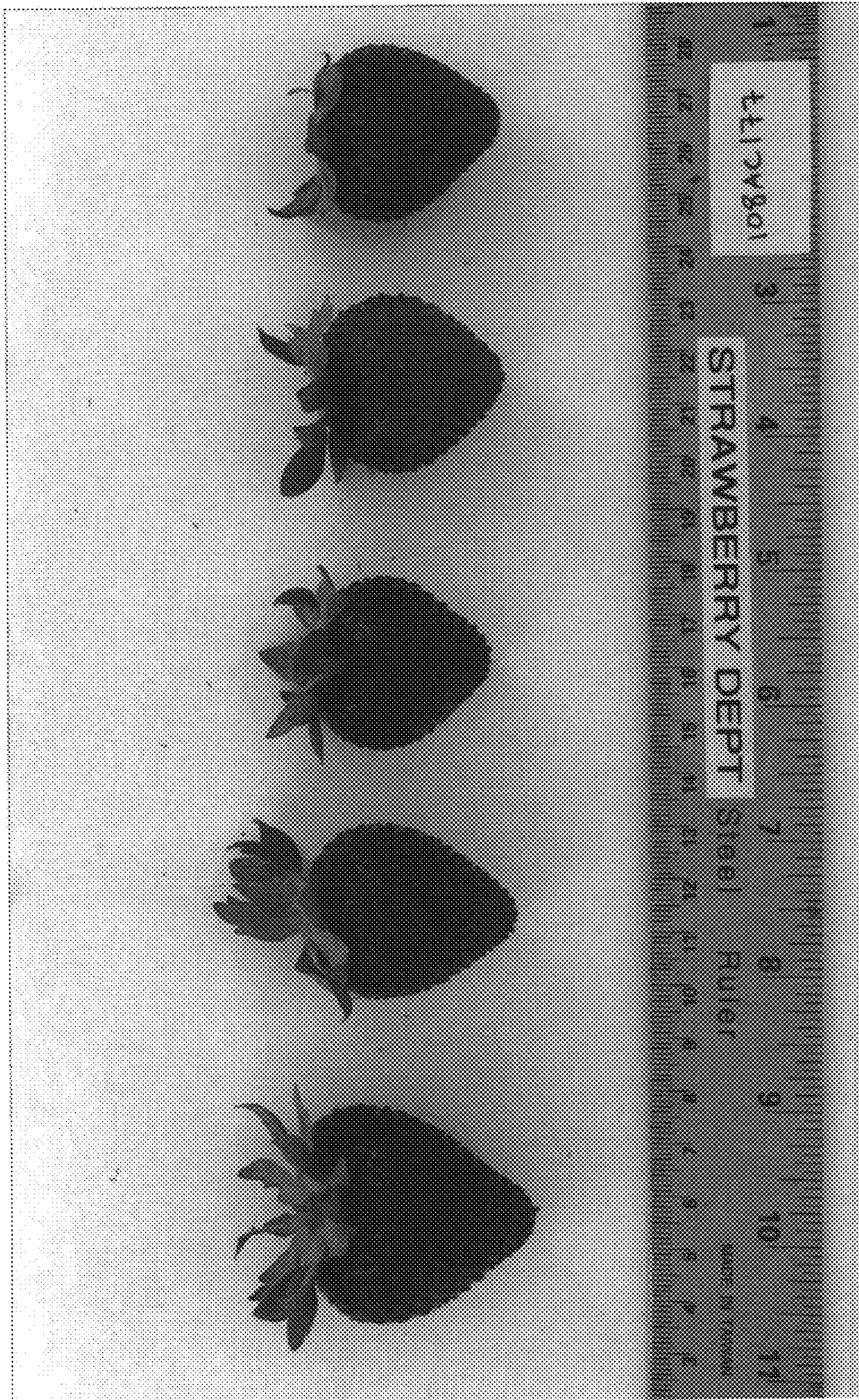


FIG. 1

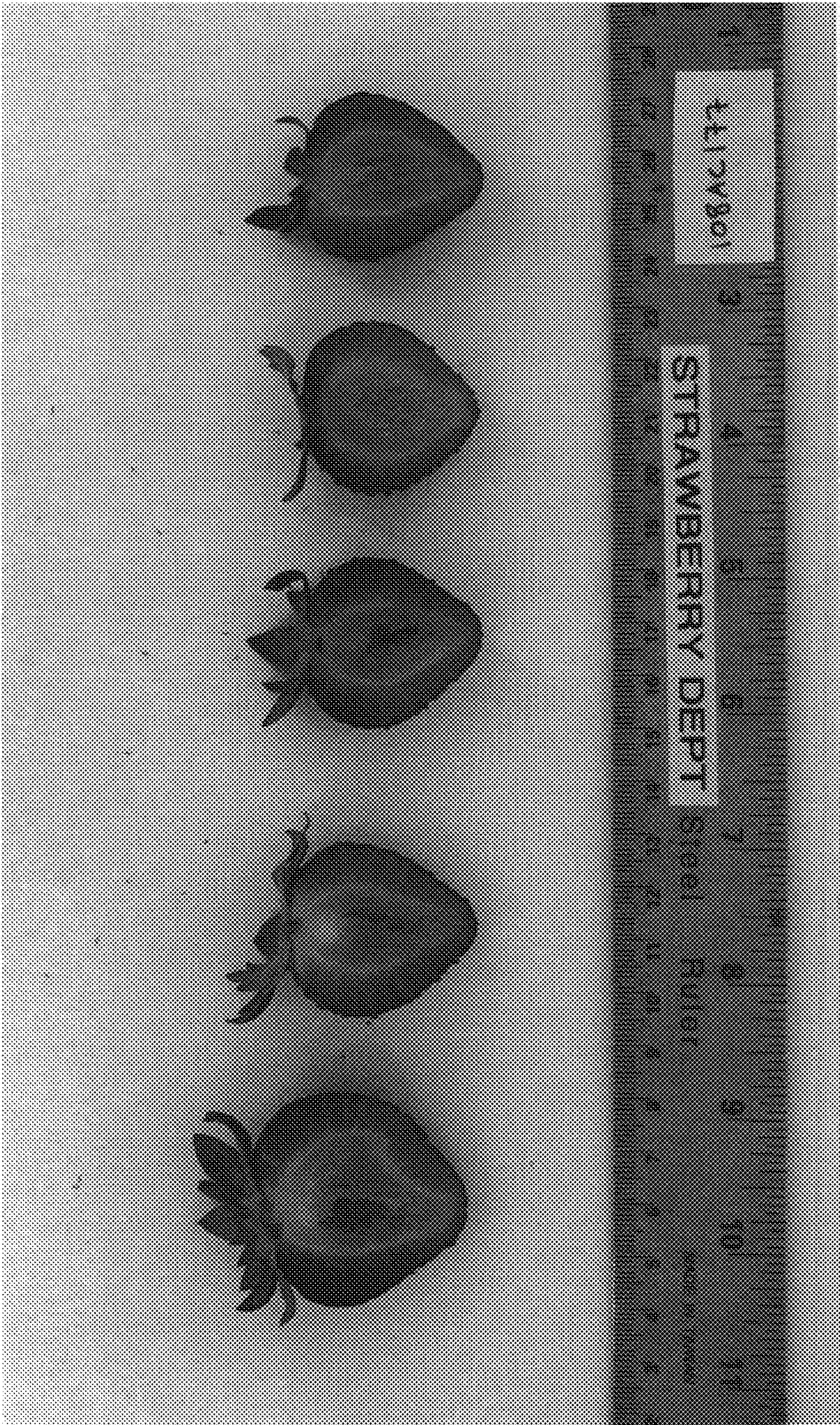


FIG. 2

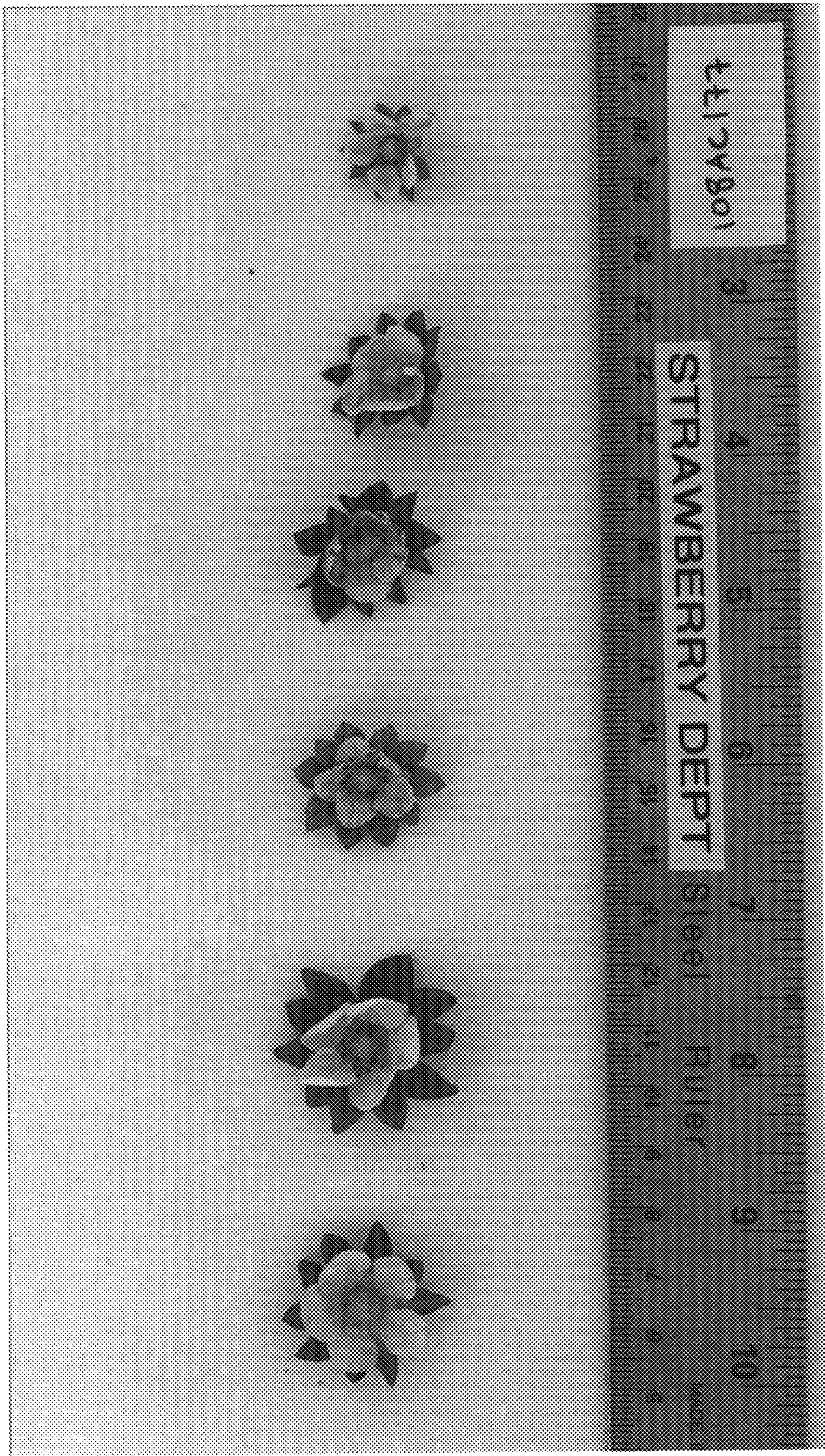


FIG. 3

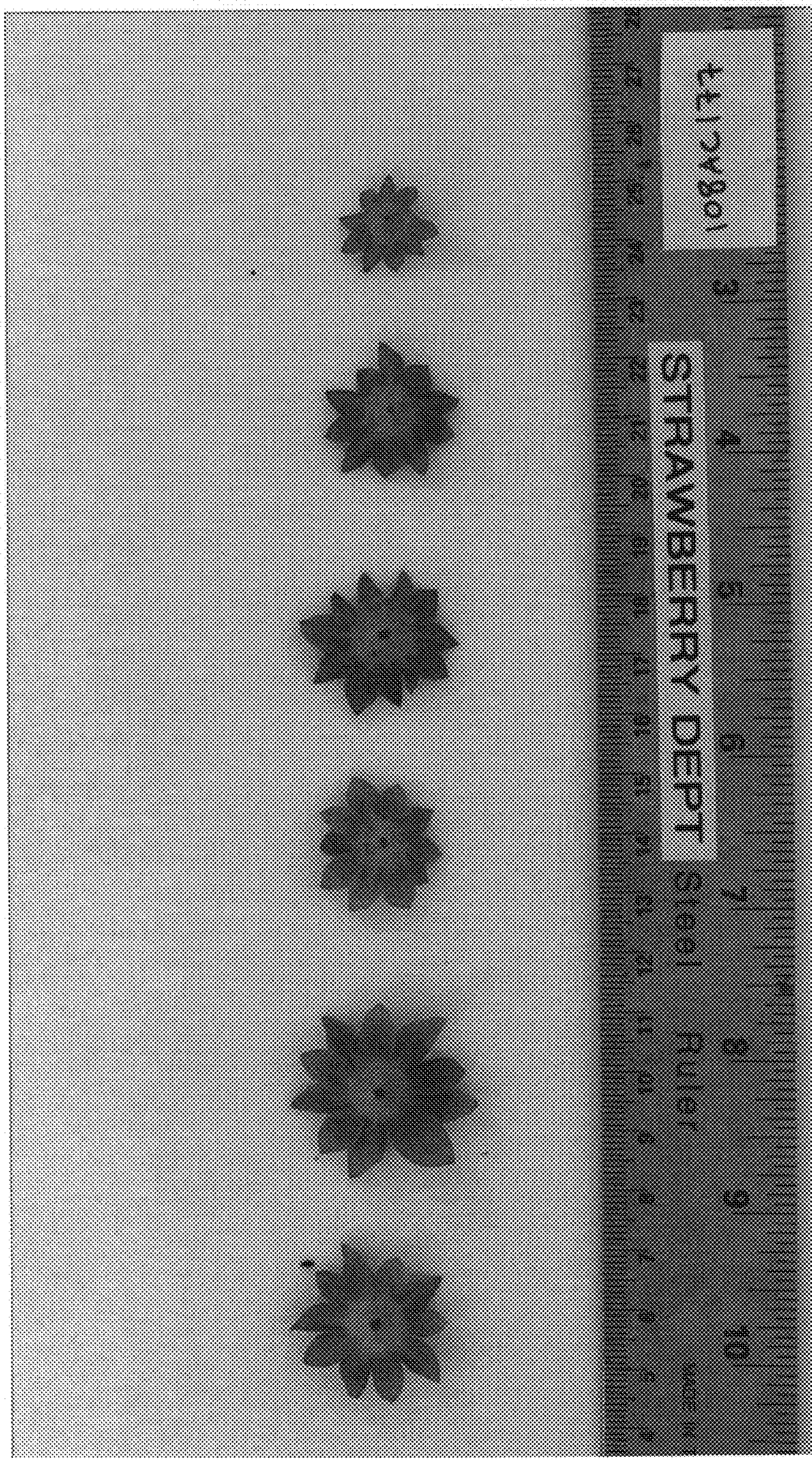


FIG. 4

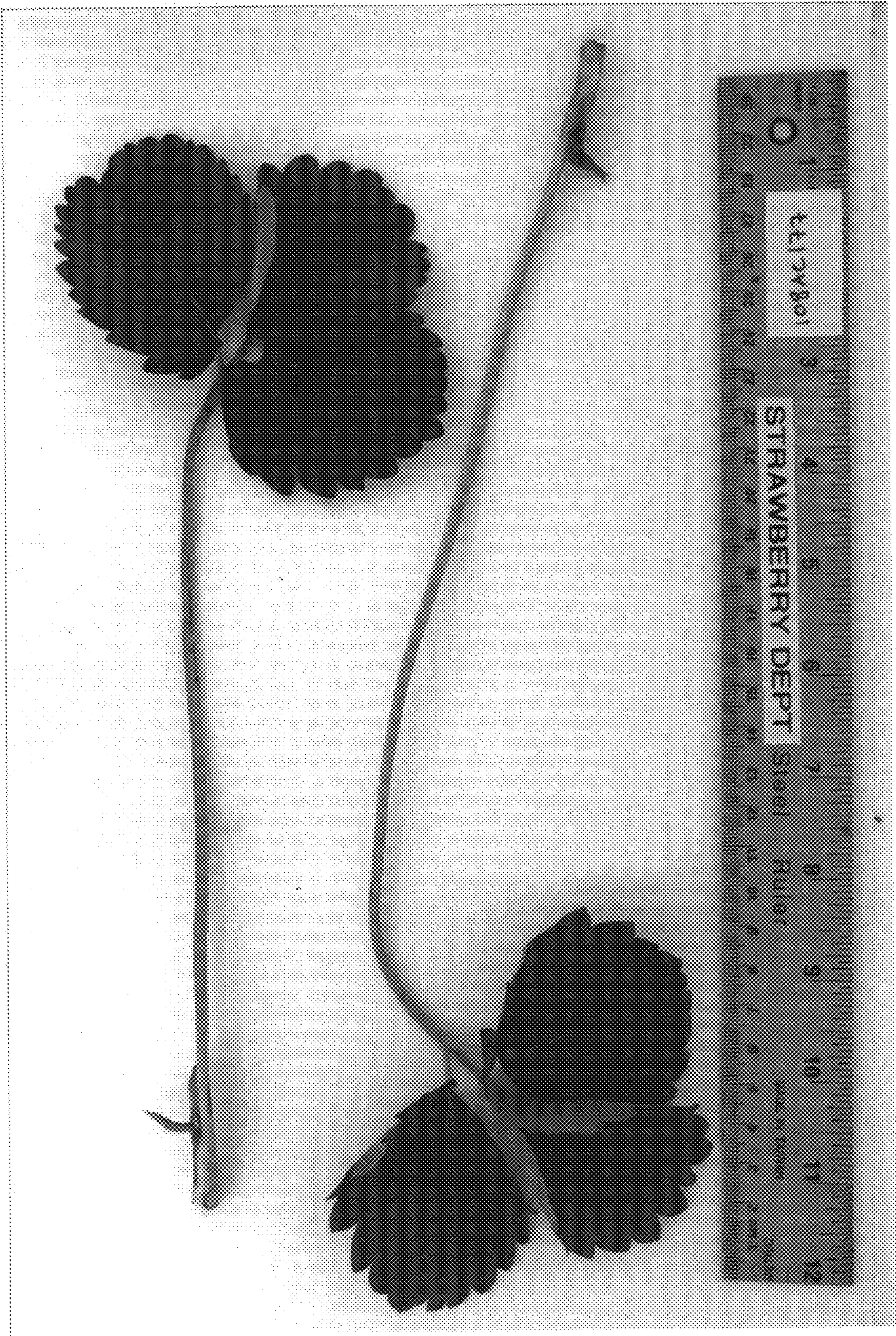


FIG. 5



FIG. 6



FIG. 7